COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

7 Data Science (djw1005)

(*a*) Let x_t be the number of new COVID infections on date t. We anticipate approximately exponential growth or decay, $x_{t+1} \approx (1+\lambda)x_t$, and we would like to estimate λ from a dataset $(x_1, \ldots, x_T)$.

(*i*) Find the maximum likelihood estimator for λ for the model

$$X_{t+1} \sim \text{Poisson}\big((1+\lambda)x_t\big)$$

[2 marks]

(*ii*) Find the maximum likelihood estimator for λ for the model

$$X_{t+1} \sim \text{Normal}\big((1+\lambda)x_t,\ (\sigma x_t)^2\big)$$

[3 marks]

(*iii*) For the latter model, explain how to compute a 95% confidence interval for λ. Explain the resampling step carefully. [4 marks]

(*b*) We don't actually know the number of new infections x_t on date t: we only know the number of new positive test results, y_t. We anticipate $y_t \approx \beta_{\text{dow}(t)} x_t$, where $\text{dow}(t)$ gives the day of the week for date t. We would like to estimate not only λ but also $\beta_{\text{Mon}}, \ldots, \beta_{\text{Sun}}$ from the dataset $(y_1, \ldots, y_T)$.

(*i*) Propose a probability model for Y_{t+1} in terms of y_t. [5 marks]

(*ii*) Explain briefly how to estimate the parameters of your model. In your answer, you should consider whether or not the parameters are identifiable. [6 marks]

Answer:

Part $(a)(i)$. It's easiest to rewrite the model as $Z_t = X_{t+1} \sim \text{Poisson}(\mu x_t)$, use example sheet 1 question 6 to write down the answer $\hat{\mu} = \sum_t z_t / \sum_t x_t$, and then say $1 + \hat{\lambda} = \hat{\mu}$ (formally, this is using the plug-in principle). But it can equally well be solved by explicitly writing out the likelihood for the original model and just maximizing it with respect to λ.

Part $(a)(ii)$ This can be approached in the standard way (write out log likelihood, maximize it), but it's better to write the model as $Z_t = X_{t+1}/x_t \sim N(\mu, \sigma^2)$, get the standard answer for $\hat{\mu}$ and $\hat{\sigma}$ from lecture notes exercise 1.9, then use the plug-in principle.

Part $(a)(iii)$. The general procedure is standard: (1) fit the unknown parameters, (2) use them to simulate a dataset X^*, (3) get an estimate $\hat{\lambda}(X^*)$ from the simulated dataset, (4) repeat many times to get a histogram for $\hat{\lambda}(X^*)$, (5) read off the 95% quantile range from the histogram. What's interesting here is how to resample. Should we simulate the entire process starting from x_0, or should we forget that the data comes from a random process and just treat each observation as independent? Either answer is acceptable, but I do need to see some reasoning.

Part $(b)(i)$ There are many possible answers, but the simplest is to just subtitute $y_t = \beta_{d_t} x_t$ into one of the two models above. For example,

$$\frac{Y_{t+1}}{\beta_{d_{t+1}}} \sim \text{Normal}\left((1+\lambda)\frac{y_t}{\beta_{d_t}},\ \sigma^2 \frac{y_t^2}{\beta_{d_t}^2} \right)$$

Rearranging makes it easier to answer the next part:

$$Z_t = \frac{Y_{t+1}}{y_t} \sim \text{Normal}\left(\frac{(1+\lambda)\beta_{d_{t+1}}}{\beta_{d_t}},\ \frac{\sigma^2 \beta_{d_{t+1}}^2}{\beta_{d_t}^2} \right)$$

Part $(b)(ii)$. The parameters of this model are not identifiable. We could scale all the β by a constant, and it would not affect the model distribution at all. We could for example fix $\beta_{\text{Mon}} = 1$. The remaining parameters can then be estimated using numerical maximum likelihood estimation. (The variance term varies, so it's not suitable for direct application of least squares.)